석사학위논문

Navigating the Accuracy-Efficiency Trade-off In Financial LLMs: A Data Centric Approach

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Navigating the Accuracy-Efficiency Trade-off in Financial LLMs: A Data-Centric Approach

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Abstract

Fine-tuning Large Language Models (LLMs) for the specialized financial domain is challenging because of the risk of overfitting and the strict constraints of the industry.1 To address these fine-tuning challenges, we propose augmenting the training dataset with diverse tasks. This data-centric approach serves as a foundation for exploring the trade-offs between model performance and computational efficiency, which are critical in finance because of cost, latency, and privacy concerns. Our research demonstrates that a primary cause of poor performance is often the lack of data diversity. An initial attempt to fine-tune a 7B parameter model on a single financial dataset (FinQA) resulted in severe overfitting and stagnant 1.6% accuracy. By augmenting the dataset, we successfully mitigated this issue and improved the accuracy to 18.06%. Based on this stable baseline, we systematically compared various optimization techniques. Our findings reveal a clear trade-off landscape, identifying 16-bit Low-Rank Adaptation (LoRA) as optimal for accuracy, Progressive Layer Dropping for a balance of performance and efficiency, and 4-bit Quantized LoRA (QLoRA) for maximum compression. This study contributes an empirical framework for selecting optimization strategies for financial applications.

- Keywords : Large Language Models, Financial AI, Data-Centric Optimization, Model Compression

I. Introduction

1. The Challenge of Domain Adaptation for Financial LLMs

The financial sector, characterized by its complex and vast data, presents significant opportunities for the application of Large Language Models (LLMs).[1] These models can enhance tasks from sentiment analysis to automated report processing.[1] However, general-purpose LLMs often lack the specialized knowledge required for high-stakes financial applications, making fine-tuning on domain-specific data a critical adaptation step.[2] Although the total volume of financial data is vast, publicly available datasets for fine-tuning are often narrow in scope and target specific tasks. This can lead to a lack of task diversity, posing a significant challenge for robust model adaptation.

Unfortunately, this adaptation process is not trivial. The financial industry demands not only high accuracy but also operates under stringent constraints related to cost, latency, data privacy, and regulatory compliance.[3] The massive computational resources required by LLMs lead to high infrastructure costs and potential latency issues, making the trade-off between performance and efficiency a central challenge for practical deployment in finance.[4]

2. The Overfitting Problem in Specialized Fine-Tuning

A significant and well-documented challenge in the adaptation process is overfitting, where a model becomes too specialized to its training data and fails to generalize to new, unseen scenarios.[5] This issue is particularly prevalent when fine-tuning high-capacity models on specialized datasets that, while high-quality, may lack sufficient task diversity.[5] Recognizing these challenges, our study aims to explore an alternative approach focused on the quality and diversity of training data.

This study empirically analyzes the impact of data diversity on model performance when applying a powerful pretrained model to the high-precision financial domain. It quantitatively compares the trade-offs of various optimization techniques (LoRA, QLoRA, Progressive Layer Dropping) under the conflicting goals of accuracy and efficiency, and presents a goal-oriented model selection framework.

3. Our Contributions

This study makes the following contributions.

1. It empirically demonstrates that data diversity is a primary factor in mitigating overfitting during the fine-tuning of LLMs for specialized

domains.

2. It provides a systematic, comparative analysis of leading optimization techniques (Full Fine-Tuning, LoRA, QLoRA, Progressive Layer Dropping) on a stable financial benchmark.

3. It proposes a practical, goal-oriented framework for selecting the optimal optimization strategy based on deployment needs, such as maximizing performance, balancing efficiency, or enabling extreme lightweighting.

II. Related Work

1. Adaptation and Lightweighting of LLMs in Finance

Adapting pre-trained models for the financial domain has become a promising approach for specialized tasks.[2] However, this domain's practical constraints necessitate Model Compression, also known as model lightweighting.[4] This field of techniques aims to reduce the size and computational complexity of models, making them more efficient for deployment.[6] In finance, this is crucial for managing massive computational costs, reducing latency in time-sensitive applications, and enabling on-premise or on-device deployment to address data privacy and security risks.[7] Research has thus begun to explore Parameter-Efficient Fine-Tuning (PEFT) methods like LoRA specifically for financial tasks, demonstrating their potential to balance accuracy with computational feasibility.[4]

2. Parameter-Efficient Fine-Tuning (PEFT)

To mitigate the prohibitive cost of full fine-tuning, PEFT methods update only a small fraction of a model's parameters.[8] Low-Rank Adaptation (LoRA) is a leading PEFT technique that freezes pre-trained model weights and injects small, trainable low-rank 'adapter' matrices into each layer. This approach drastically reduces trainable parameters and memory requirements.[9] Quantized LoRA (QLoRA) further improves efficiency by quantizing the base model to a lower precision, such as 4-bit, while still training the LoRA adapters. Its key innovation is the 4-bit NormalFloat (NF4) data type, an information-theoretically optimal format for the typical distribution of neural network weights, which preserves more information than standard integer quantization.[10]

3. Architectural Pruning: Progressive Layer Dropping

A more aggressive approach to model compression is architectural pruning, such as Progressive Layer Dropping, which systematically removes entire layers from a transformer model.[11] Research has shown that not all layers contribute equally to performance on a specific task.[12] Methods like TrimLLM combine fine-tuning with progressive layer removal, demonstrating that LLMs can be compressed to less than 50% of their original size for domain-specific tasks with negligible accuracy loss.[11] A key advantage of this technique is that it reduces inference latency without requiring specialized hardware, making it a practical method for resource-constrained environments and a useful tool for investigating knowledge distribution within the model.[11]

4. Knowledge Distribution in Transformer Architectures

A fundamental question in LLM research is how knowledge is stored in the network. Some studies suggest a functional specialization, where early layers handle basic linguistic features, middle layers manage higher-level concepts, and final layers perform task-specific reasoning.[12] Other work has identified localized "knowledge circuits" for specific facts.[13] However, architectural pruning studies often find that complex reasoning is more diffusely distributed, as dropping a significant number of layers leads to graceful, rather than catastrophic, performance degradation.[14] This debate provides the context for this paper's argument that financial reasoning is a distributed, emergent property of the entire network.

III. Preliminary Study: Limitations of Fine-tuning on a Single Dataset

This section presents a preliminary study that forms the problem statement for our research. We demonstrate the limitations of fine-tuning a large-capacity model on a single, narrow dataset, which motivates our proposed data-centric methodology. All experiments utilized the Llama-2-7B model, which was chosen because it is a powerful, open-source, and widely used foundational model that serves as a strong and reproducible baseline for academic research.[15]

Our initial experiment was designed to fine-tune the Llama-2-7B model on the FinQA dataset, which is a high-quality benchmark for numerical reasoning in financial reports.[16] Despite the quality of the dataset, its focus is narrowly confined to quantitative reasoning. When the high-capacity Llama-2-7B model was trained on this single-task dataset, it exhibited severe overfitting. The model memorized the specific patterns within FinQA rather than learning generalizable financial reasoning principles, which led to a performance collapse. On a held-out test set, the model achieved a stagnant execution accuracy of only 1.6%, demonstrating its failure to adapt effectively. This result highlights that a high-quality but narrow dataset is insufficient to properly fine-tune a large model for a complex domain.

IV. Proposed Methodology: Data-Centric Optimization via Diversified Corpus

1. Data-Centric Approach

To address the overfitting problem identified in our preliminary study, we proposed a data-centric solution, which prioritizes the systematic improvement of data quality over model-centric tuning. This approach posits that for many real-world problems, engineering a more representative and diverse dataset is a more effective path to better performance than iteratively tweaking the model architecture.[17]

2. Corpus Augmentation via Heterogeneous Datasets

Our core methodology involves augmenting the training corpus to enhance task diversity. We engineered a new diversified corpus by strategically combining three distinct datasets, each contributing a unique reasoning modality:

- FinQA: Provides foundational training in numerical reasoning, requiring mathematical operations over figures found in financial tables and text.
- TAT-QA: Introduces more complex hybrid reasoning by requiring the synthesis of information from both tables and multiple paragraphs of surrounding text in real-world financial reports.[18]
- FiQA: Adds the critical element of qualitative reasoning by focusing on opinion-based question answering and sentiment analysis from unstructured sources such as news, microblogs, and forums.[19]

By combining these datasets, we created a comprehensive training curriculum that forced the model to develop a more holistic and robust understanding of financial language, as it had to learn representations useful for quantitative, hybrid, and qualitative tasks simultaneously.

V. Experiments and Results

1. Validation of the Data-Centric Approach

To validate our proposed method, we performed full parameter fine-tuning of the Llama-2-7B model on the newly created augmented corpus. The impact of the data-centric approach was immediate and profound in the following ways. As shown in <Table 1>, the execution accuracy of the model increased from 1.6% to 18.06%. This greater than tenfold improvement successfully resolved the initial overfitting issue and established a stable, meaningful baseline. This result confirms that data diversity is a critical prerequisite for successfully adapting LLMs to the financial domain and provides a solid foundation for the subsequent comparative analysis of the optimization strategies.[20]

2. Further Exploration: Comparative Analysis of Optimization Strategies

With a stable baseline performance established, we designed a second set of experiments to systematically explore the trade-off between accuracy and efficiency of the model. This analysis is motivated by the practical needs of the financial industry, where model compression is crucial for managing costs, reducing latency, and enabling privacy-preserving on-premise deployment.[2] We implemented and compared four distinct optimization strategies against our full fine-tuning baseline.

- 16-bit LoRA: A PEFT method to reduce trainable parameters and memory usage.
- 4-bit and 8-bit QLoRA: More aggressive compression techniques combining PEFT with weight quantization.
- Progressive Layer Dropping: An architectural pruning technique in which entire layers are removed post-fine-tuning. Based on layer importance probing, we removed the six least important initial layers in sequence to strategically reduce the model depth.

All models were trained on the same augmented financial corpus and evaluated using execution accuracy on a held-out dataset.

3. Comparative Analysis Results

The results of our comparative analysis are shown in <Table 1>. They revealed a clear trade-off between model accuracy and computational efficiency.

<Table 1>: Comparative Performance of Optimization Strategies.

Optimization Method	Final Accuracy (%)	Performance vs. Baseline (%)	Model Size Reduction (%)
FinQA-only (Initial Attempt)	1.60%	8.90%	0%
Full Fine-Tuning (Baseline)	18.06%	100%	0%
16-bit LoRA	18.08%	100.10%	N/A
Progressive Layer Dropping (6 Layers)	16.23%	89.90%	18.75%
4-bit QLoRA	9.13%	50.50%	~75%
8-bit LoRA	8.96%	49.60%	~50%

- The 16-bit LoRA slightly exceeded the performance of the fully fine-tuned baseline, suggesting that its regularization effect led to better generalization.
- Progressive Layer Dropping provided the most compelling balance, retaining nearly 90% of the baseline performance with a

significant 18.75% reduction in model depth.

- The quantization results were particularly revealing in this regard. The 4-bit QLoRA achieved a higher accuracy (9.13%) than the 8-bit LoRA (8.96%), underscoring the importance of the quantization methodology.

VI. Discussion

1. Data Diversity as the Primary Determinant of Success

The profound success of the diversified corpus validates the data-centric AI paradigm.[17] The combination of quantitative (FinQA, TAT-QA) and qualitative (FiQA) tasks acts as a form of implicit regularization. To minimize loss across this combined objective, the model was compelled to develop internal representations of financial concepts that were robust enough for both calculation and interpretation. This demonstrates that investing in data diversity can yield performance gains an order of magnitude greater than those from model-centric tuning alone in a specialized domain.

2. Navigating the Accuracy-Efficiency Trade-off Landscape

The results show that the optimal optimization choice is contingent on the specific deployment goal, allowing for a pragmatic framework with clear real-world financial applications:

- Goal: Maximum Performance. For applications where accuracy is paramount, a 16-bit LoRA is the superior choice. Its ability to slightly outperform full fine-tuning makes it ideal for high-stakes tasks, such as real-time fraud detection or algorithmic trading, where even fractional performance gains are critical.
- Goal: Balanced Performance and Efficiency. For many enterprise applications, Progressive Layer Dropping represents the optimal equilibrium. Sacrificing a marginal ~10% of performance for a significant ~19% reduction in model size and latency is a highly favorable trade-off for tasks like internal market analysis report generation or automated credit risk assessment, where both responsiveness and high accuracy are important.
- Goal: Extreme Lightweighting. For deployment in highly resource-constrained environments, 4-bit QLoRA is the undisputed winner. Its ~75% reduction in memory footprint makes deployment feasible for on-device applications, such as personalized product recommendations within a mobile

banking app, or on local terminals for privacy-sensitive tasks that cannot rely on cloud APIs.

3. The Superiority of Information-Theoretic Quantization

The outperformance of 4-bit QLoRA over 8-bit LoRA (9.13% vs. 8.96% accuracy) is a powerful demonstration that the *method* of quantization is more critical than bit-depth alone. Standard 8-bit integer quantization inefficiently maps values for the normal distribution of neural network weights. In contrast, QLoRA's 4-bit NormalFloat (NF4) data type is information-theoretically optimal for this distribution, as its quantization bins are placed to represent equal portions of the probability mass. This allows NF4 to preserve more salient information in 4 bits than a naive uniform scheme does in 8 bits, explaining its superior performance.[10]

4. Financial Reasoning as a Distributed Function

The graceful degradation in the Progressive Layer Dropping experiment suggests that financial reasoning is not localized but is a distributed, emergent property of the network's depth. This aligns with literature suggesting that for complex tasks, capabilities are spread across the architecture. Financial QA requires integrating semantic parsing, information identification, and mathematical calculation. It is plausible that each layer incrementally refines the input representation, contributing a piece to this complex reasoning chain.[22] Removing the identified low-importance layers reduces the model's overall "reasoning capacity" rather than eliminating a single, critical function, leading to a proportional, not precipitous, decline in performance.

VII. Conclusion

1. Summary of Findings

This study empirically analyzed the impact of data diversity on fine-tuning LLMs for the financial domain and quantitatively compared various optimization techniques. Our conclusions are summarized by three core arguments:

1. Data diversity is the paramount factor for success. A data-centric approach resolved catastrophic overfitting and improved accuracy by over an order of magnitude.
2. The optimal optimization strategy is goal-dependent. A clear framework was established: 16-bit LoRA for maximum performance, Progressive Layer Dropping for balance, and 4-bit QLoRA for maximum efficiency, with each being suitable for different financial applications.
3. Intelligent, distribution-aware

quantization is superior to naive methods. The information-theoretically optimal NF4 data type in 4-bit QLoRA preserved more information with fewer bits than standard 8-bit quantization.

2. Limitations and Avenues for Future Research

This study was conducted on a single base model (Llama-2-7B), and future work could investigate if these findings generalize to larger models or different architectures. This study's layer dropping strategy, while based on importance probing, was static as it removed a pre-determined number of layers. As a future research direction, an adaptive strategy that dynamically removes the most inefficient layers based on layer importance scores during the fine-tuning process could be explored. This could yield a more optimal trade-off between compression and performance. Finally, a promising direction is the combination of these techniques, such as applying QLoRA to a layer-pruned model, which could push the boundaries of efficiency even further.

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